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RESEARCH ARTICLE

Safety Helmet Wearing Detection Model Based on Improved YOLO-M

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ABSTRACT In recent years, construction accidents have occurred frequently. The safety guarantee measures for construction personnel are thrown into sharp focus. Wearing helmets is one of the most important requirements to protect the safety of construction personnel, and the detection of wearing safety helmets has become necessary. For the problems of the existing helmet wearing detection algorithm such as too many parameters, substantial detection interferences, and low detection accuracy, in this paper a helmet wearing detection model YOLO-M is proposed. Firstly, MobileNetv3 is adopted as the backbone network of YOLOv5s for the feature extraction, which can reduce the number of model parameters and model size. Secondly, a residual edge is introduced in the feature fusion. The original feature map information is fused during feature fusion, and the detection ability of small targets is enhanced. At last, by changing the connection between CAM and SAM, a new attention module BiCAM is designed. The comparison experiments show that the detection accuracy of YOLO-M is 2.22% higher than YOLOv5s, and the model parameter quantity is reduced to 3/4 of YOLOv5s. Under the same detection conditions, the detection speed of YOLO-M is better than the other models, which meets the accuracy requirements of helmet detection in the construction scene.

INDEX TERMS Attention mechanism, feature fusion, safety helmet, YOLOv5s model.

I. INTRODUCTION

With the development of social informatization, intelligent devices are widely used in the industries. For example, the license plate recognition system is used in parking lots, and the face recognition system is used in consumer places, etc. In the construction industry, the construction environment is complex, and the injury risk of falling objects is high. Safety helmets can protect the workers and reduce the impact of falling objects, and effectively prevent workers from being injured by falling objects [1]. Workers who do not wear safety helmets have great potential safety hazards. Therefore, wearing safety helmets in production operations can significantly protect the safety of the workers' lives. Safety helmet wearing detection is essential for the safety of workers. In the early days of construction sites, some people were specially designated to supervise whether workers wearing safety helmets or not. However, construction workers worked in a wide range and it was difficult to find the workers who did not wear the safety helmets. In addition, the traditional supervision method wasted the manpower [2]. For the safety production requirements on contemporary construction sites, deep-learning-based computer vision methods are more suitable for real-time monitoring of construction sites.

With the fast research development in the field of deep learning, many researchers began to use deep learning algorithms on the safety-helmet detection. In the helmet detection algorithm based on YOLO, the output dimension of the classifier was modified to reduce the numbers of parameters. The algorithms had good real-time performance but the accuracy was low [3]. An improved target detection algorithm based on YOLOv3 combined the advantages of IoU

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and GIoU, and it focused on overlapping areas and non-overlapping areas. Furthermore, a new objective function was proposed, which improved the accuracy [4]. An algorithm applied the lightweight structure on the MobileNet network and it compressed YOLOv2 [5]. YOLO-S was a lightweight helmet wearing detection model. It balanced data categories by data enhancement, and replaced the original backbone network with the lightweight network MobileNetV2. It reduced the network calculation, and finally performed pruning and knowledge distillation on the model channel [6]. A real-time helmet wearing detection algorithm embedded the ECA-net module into YOLOX's feature fusion network, and used CIOU loss instead of DIOU loss to obtain better helmet detection results [7]. Light-YOLOv4 uses sparse training and pruning strategies based on YOLOv4. The improved model compresses the model size to 10% with a slight loss of accuracy, and the detection speed is also greatly improved [8]. BiFA-YOLO proposes a two-way feature interaction method, which effectively combines multi-scale feature information. Aiming at the problem of inconsistent target angles, this paper proposes an angle classification structure to obtain target angle information [9]. Although the above methods improved the performance of the detection, there are still some disadvantages. For example, the two-stage algorithm models are large, the training time is long, and they require more system resources. Due to the large scale of the models, the deployment is difficult. Similarly, the robustness of the single-stage models is weak. In addition, they still have a high missed rate and false rate for long-distance small targets. It is necessary to further improve the performance to cope with complex and dense scenes.

A. TWO KINDS OF OBJECT DETECTION

The current mainstream object detection models are mainly divided into two categories, one-stage network and two-stage network. The difference between them is whether a separate network is required to generate candidate boxes. Region-CNN (RCNN) is one of the representative works of two-stage detection network. The target images extracted by the region proposal was normalized and input into CNN. However, due to a large number of overlaps in the anchor boxes, the computing resources were wasted [10]. Instead of convolving each anchor box, Fast RCNN performed one convolution on the whole image, which simplified a lot of repeated convolution calculations. Then a RoI pooling layer was added after the convolution to generate a fixed-size feature map. It accepted input features of any size [11]. Faster RCNN creatively used RPN instead of the original selective search method to generate anchor boxes, and reduced the number of anchor boxes from the 2000 to 300, which improved the detection speed of the network [12]. The model was verified on the PASCAL VOC 2007 dataset [13], and the experiment showed that the detection accuracy of the model was gradually improved. It means that the extract feature method had gradually changed from selective search to region proposal network (RPN).

The one-stage detection network YOLO detects faster. This network model defines the detection task as a regression, which avoids the dividing the detection task into two steps in the R-CNN series network [14].

B. CONTRIBUTIONS

According to the above problems, this paper chooses the typical one-stage detection model YOLOv5 as the baseline to improve the algorithm. In the one-stage models, there is no process of generating prediction boxes. So compared with the two-stage network, it has great advantages in the detection speed and network size. In the typical single-stage models, the SSD model is a lightweight model, but it uses VGG16 as the backbone network and lacks the multi-scale feature fusion. So the detection effect is not satisfactory [15]. YOLOv5 inherits the advantages of the YOLO series of models, and it also brings about some improvements such as MOSIC data enhancement, expanding input channels by focusing structure, increasing positive sample matching, etc. However, for the helmet detection task of the target construction site studied in this paper, YOLOv5 still has some shortcomings. Its backbone network is too large to deploy on a low-computing platform. In addition, the model does not fuse the global information and the context information well. In the dense helmet object detection, the performance needs to be improved.

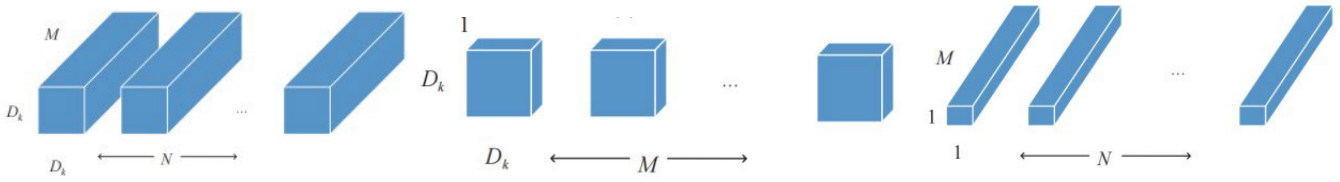
According to the above analysis, based on YOLOv5s model, this paper makes some improvements and proposes a YOLO-M (YOLO Mini) helmet detection model. YOLO-M is an improvement on the application of helmet wearing detection, and it is a lightweight helmet detection model. The core idea of the algorithm is to compress the model and improve the detection effect of helmets. On the basis of ensuring the accuracy of the model, YOLOv5s is modified, and the features of the target dense area of the detected images are mined.

The YOLO-M model proposed in this paper takes into account the shortcomings of the YOLO model in the helmet detection task, and makes three main contributions as follows:

(1) In terms of feature extraction network, considering that CSPDarknet53 has too many parameters, this method chooses the lightweight network MibileNetv3 as the backbone network. It is connected to the feature fusion module by adjusting the output feature map size of MibileNetv3.

(2) The BiCAM attention module is designed. By improving the combination of spatial attention and channel attention, this method obtains attention weights based on two dimensions of space and channel. BiCAM is embedded into the model to enhance the features of dense objects and reduce unfavorable background interference.

(3) Referring to the structure of the residual network, a new multi-scale feature fusion module Res-FPN is proposed. Combined with BiCAM, it obtains more effective



(a) Standard convolution (b) Depthwise convolution (c) Pointwise convolution

FIGURE 1. Standard convolution decomposition process.

attention weights of the feature maps. It improves the ability of the model to obtain the global information of feature maps and enrich context information. It allows the model to pay more attention to the tiny and dense helmet targets during each fusion pass. This method improves the efficiency of the feature fusion, enhances the recognition accuracy of the model in general scenes, and optimizes the helmet detection ability of dense targets in occluded scenes.

II. PRELIMINARY KNOWLEDGE

A. DEPTHWISE SEPARABLE CONVOLUTION

The MobileNet series of networks are representatives of lightweight networks. Depthwise separable convolutions are introduced to the networks, and they are widely used in mobile terminals. The standard convolution is multiplied and accumulated on the feature map channel by a convolution kernel with the same number of channels as the input data. The depthwise separable convolution mainly consists of two steps, depthwise convolution and pointwise convolution. Fig. 1 shows the standard convolution decomposition process.

In Fig. 1, $\boldsymbol{D_K} \times \boldsymbol{D_K}$ represents the size of the convolution kernel, M represents the number of input channels, and N represents the number of output channels.

Let $\boldsymbol{D_F} \times \boldsymbol{D_F}$ be the size of the input feature, and then the computational cost of $\boldsymbol{N}$ convolution kernels on a fully convolutional layer can be computed as follows:

$$\mathrm{Cos}_{full} = D_K \times D_K \times N \times M \times D_F \times D_F \tag{1}$$

The calculation cost of 1×1 pointwise convolution is as follows:

$$\mathrm{Cos}_{1\times1} = N \times M \times D_F \times D_F \tag{2}$$

The calculation cost of depthwise separable convolution is as follows:

$$\mathrm{Cos}_{all} = \mathrm{Cos}_{full} + \mathrm{Cos}_{1\times1} \tag{3}$$

From (1) and (3), the ratio of the depthwise separable convolution calculation to the standard convolution calculation is as follows:

$$\frac{Cos_{all}}{Cos_{full}} = \frac{D_k \times D_K \times M \times D_F \times D_F + N \times M \times D_F \times D_F}{D_K \times D_K \times N \times M \times D_F \times D_F} = \frac{1}{N} + \frac{1}{D_K^2} \tag{4}$$

Therefore, if the convolution is 3×3, the number of convolution kernels increases, and the effect of the depthwise separable convolution is better. If the same number of convolution kernels is used for the standard convolution, the ratio of the calculation amount of the deep separable convolution to the standard convolution is 1/9 at most.

B. MOBILENETV3

On the basis of deep separability and the reverse residual module of linear bottleneck, the SE module is introduced in MobileNetv3 [16], [17], and it is also be added to the bottlenet structure. As shown in Fig. 2, it mainly includes two steps, squeeze and excitation. W, H, and C represent the width, the height, and the number of channels of the feature map respectively. So the input feature map size is $H \times W \times C$. In the process of squeeze, the feature map is compressed into a $1 \times 1 \times C$ vector by global average pooling. The process of excitation consists of two fully connected layers, which generates a $1 \times 1 \times C$ weight vector. Finally, the input feature map is multiplied by the weight vector through the scale operation to obtain the output result.

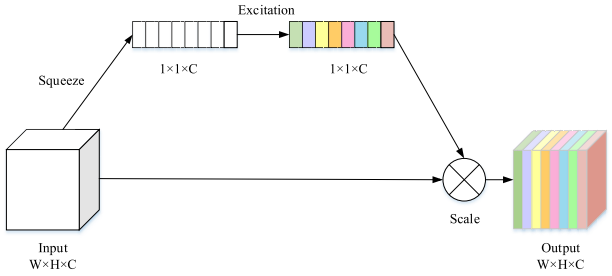


FIGURE 2. Squeeze and excitation module.

For YOLOv5, the whole network are divided into three parts, the backbone feature extraction network (Backbone), the enhanced feature extraction network (Neck), and the prediction network (Head). In our work, CSPDarknet53 is replaced by the lightweight MobileNetv3 as the backbone network for feature extraction. According to the YOLOv5 network structure, it is necessary to use the backbone network to obtain 3 effective feature layers to strengthen the construction of the feature extraction pyramid. As shown as Fig. 3, in the original YOLOv5 backbone network there are 3 CSPDarknet53 feature layers (80, 80, 256), (40, 40, 512) and (20, 20, 1024), which can be replaced by the effective

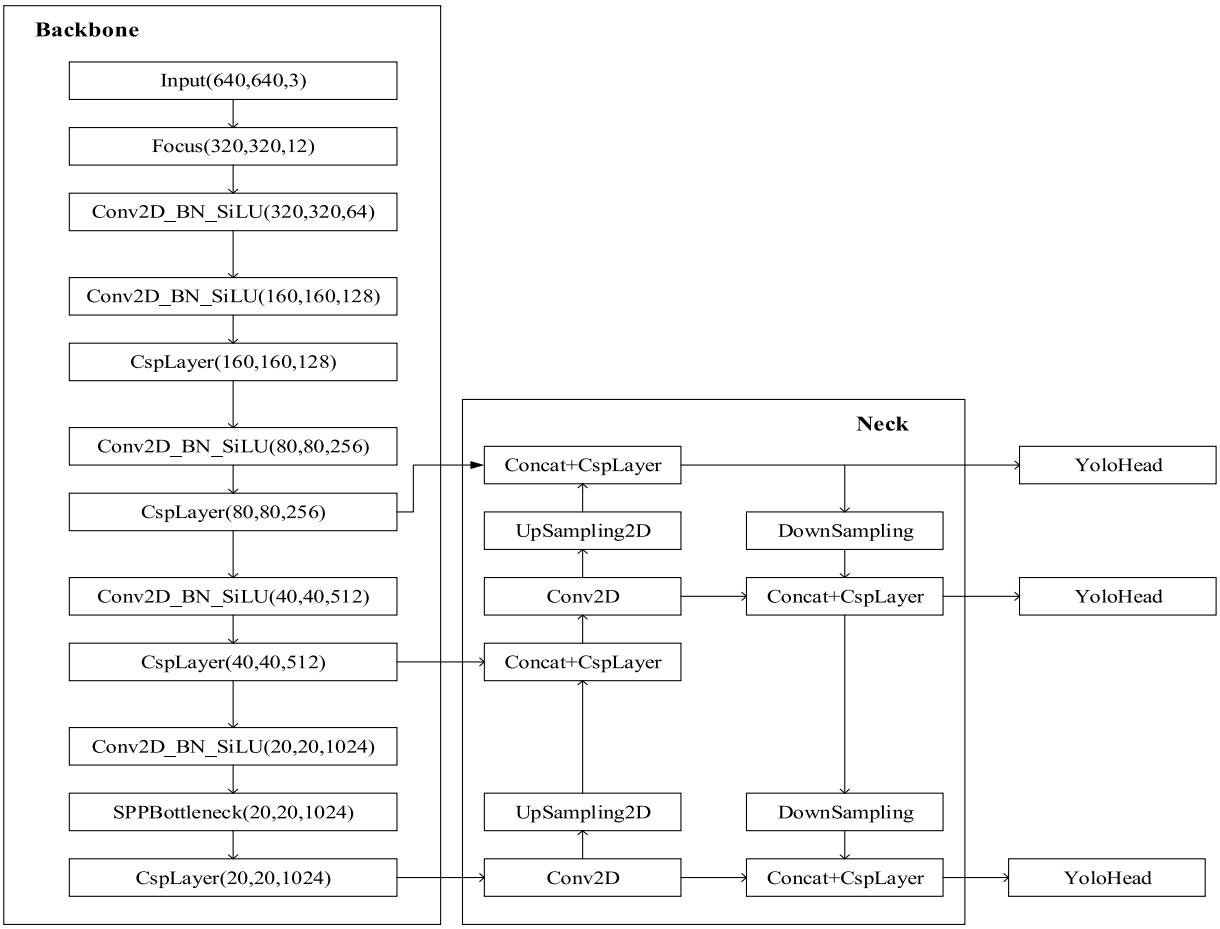


FIGURE 3. YOLOv5 network structure.

feature layer obtained from MobileNetv3. This purpose of the improvement idea is to reduce the number of the model parameters.

C. YOLOV5S MODEL

YOLO generalizes the object detection as a regression problem and realizes end-to-end training and detection. Because of good balance between speed and accuracy, it is popular in the field of target detection in recent years and has been successfully studied, improved and applied to many different fields. Reference [18] systematically reviewed the important improvements of the YOLO series network. YOLOv5 is a new achievement of the YOLO series of object detection networks. YOLOv5 has four nework models, YOLOv5s, YOLOv5m, YOLOv5l and YOLOv5x. Among them, YOLOv5s is the network with the smallest depth and the smallest width of the feature map, and the overall architecture of the networks is the same. Based on YOLOv5s, the improvement of safety helmet wearing detection is studied.

The YOLOv5s algorithm framework is shown in Fig. 3. It consists of a backbone network, a neck and a head. YOLOv5 is the follow-up work of YOLOv4. In the backbone the Focus network structure is added and the input channel is expanded by four times. Mosaic data enhancement is added to the image preprocessing, which means that four images are used for stitching. In the feature fusion structure of neck, YOLOv4 does not introduce the cross stage partial (CSP) structure, and YOLOv5s adds the CSP structure to Pan-Net. In the head, the loss function CIOU_Loss of the anchor box is replaced with GIOU_Loss [19]. Weighted non-maximum suppression (NMS) operation is adopted to filter target anchor boxes to improve the accuracy of target detection [20].

III. YOLO-M SAFETY HELMET WEARING DETECTION MODEL

A. IMPROVEMENTS OF FEATURE EXTRACTION

The module located between the backbone and the head is usually called the neck. Before feeding the features extracted by the backbone to the detection layer, it fuses more feature information. When the features of small objects are extracted to a higher dimension, this structure can effectively prevent the loss of features of small objects. It affects the detection results by up-sampling the feature map again. In this way, the feature information of different layers of the fusion backbone can be extracted and fused to affect the detection results.

Feature pyramid network (FPN) is now widely used in multi-scale feature fusion. Recently in Pan-Net [21], NAS-FPN and other methods, many structures have been studied for cross-scale feature fusion. For example, Pan-Net is adopted in YOLOv5s. The FPN structure is shown in Fig. 4(a), which is a top-down process. The top feature maps are upscaled to the same size as the shallow feature maps by upsampling and fusion. It not only exploits the abstract semantic features of high-dimensional feature maps, but also combines the features of shallow feature maps. In Pan-Net, extraction is implemented iteratively on the basis of FPN, as shown in Fig. 4(b). After completing the top-down process of the feature pyramid, bottom-up feature extraction is required, and the features of the same layer also need to be fused. All the features fusion of different scales in FPN and Pan-Net is simple addition. By default, the contributions of different scale features are equal. However, in practical applications the situations are complex. For example, when detecting the helmet wearing, the detection objects are usually complex, dense and small. In addition, multiple feature extractions cause the network degradation easily, which leads to the loss of small helmet targets in the high-dimensional features Therefore, in these networks the contribution of high-dimensional features are sometimes less than that of shallow features.

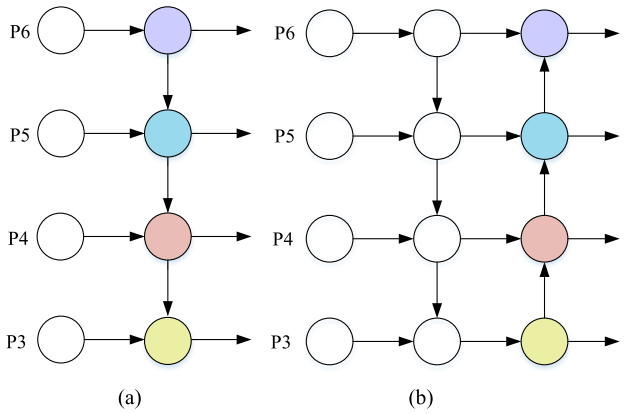


FIGURE 4. Neck structure.

The ResNet network is designed to solve the problem of network degradation and gradient disappearance,. It contains residual blocks, which applies the residual edges to combine the output features with the input features to enhance the network performance. In this paper a Res-FPN structure is designed on the base of the residual connection, which is shown in Fig. 5. When the feature fusion is performed in the same layer, the sizes of the feature maps are the same, so there is no need to perform upsampling or downsampling to adjust the sizes. The residual edge can directly connect to the original input feature map, whose global information is fused with the feature map of local information obtained after deep convolution. The operation is represented by the dotted line in Fig. 5. This structure allows the network to learn more small-scale information, and improves the accuracy of safety helmet detection.

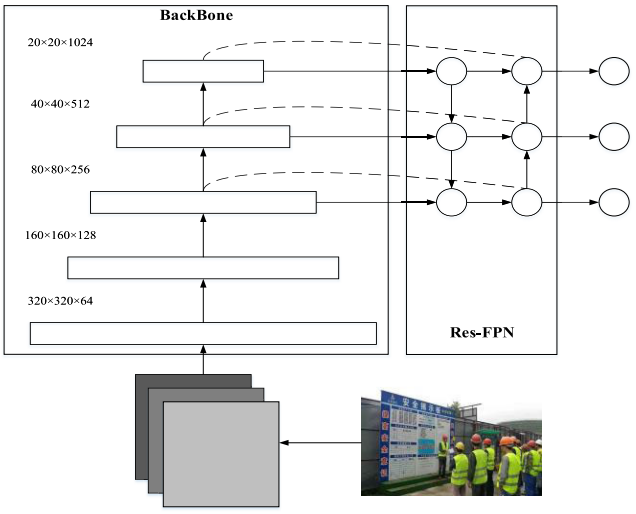


FIGURE 5. Res-FPN Feature fusion module.

B. IMPROVEMENTS OF ATTENTION MECHANISM

The attention mechanism is to process the data by searching for the key local information in the global information, which enables the model to focus on the helpful information and exclude the useless information. The attention module CBAM [22] consists of a channel attention module (CAM) and a spatial attention module (SAM), which is shown in Fig. 6. The module well combines the context feature information and weights the channels with rich semantic information to improve the detection effect. However, it also can be seen that when SAM calculating, the input features are ignored. The output of SAM is only added by the output of CAM, and the deep network loses the global information of the input feature layer through layer-by-layer convolution. It is not able to combine the global information and the local information well.

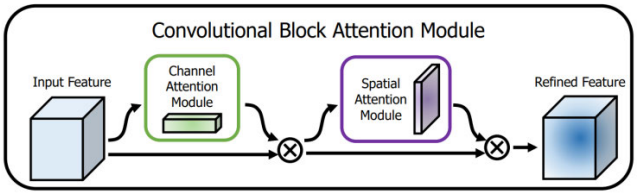


FIGURE 6. CBAM module.

Based on the above analysis, a BiCAM module is proposed, which is shown in Fig. 7.

BiCAM pays attention to the contextual information while takes the global information into consideration. In convolutional neural networks, CAM focuses on the effective feature layers. It makes the network pay attention to more useful information for detection of the images by assigning weights to the channels. For the $H \times W \times C$ input feature map Y,

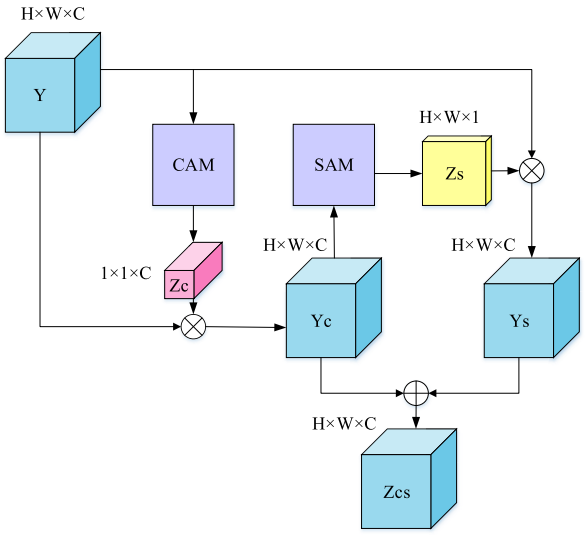


FIGURE 7. BiCAM attention module.

the channel attention module adopts average pooling and max pooling in the spatial dimension to collect the target features. After applying the two pooling operations at the same time, the size of the feature map and the amount of computation are reduced, and the expressiveness of the network is improved. The two one-dimensional vectors after pooling are sent to the fully connected layer. Finally, the channel attention Z_c is generated through element-wise summation and sigmoid activation function, which is shown in (5).

$$Z_c = \sigma(MLP\,(AvgPool\,(Y)) + MLP(MaxPool(Y))) \quad (5)$$

SAM pays more attention to the location information of the feature, and focuses on the location that contains more effective information in the feature map. It performs channel-wise compression via average pooling and max pooling to generate two feature maps. It performs the *concat* opration to get a feature map with two channels, and then convolves using a convolutional layer with a single convolution kernel. Finally, the sigmoid function is adopted to generate the feature map of the spatial attention Z_s, which is shown in (6). In the whole process, SAM and CAM complement and promote each other.

$$Z_s = \sigma(f^{7\times7}([Avgpool\,(Y)\,, Maxpool(y)])) \quad (6)$$

In (6), $f^{7\times7}$ represents a convolution operation using a 7×7 convolution kernel.

Based on CAM and SAM modules, a new BiCAM attention module is proposed. First, the channel attention weight Z_c is obtained according to CAM, and it is multiplied with the input feature Y to obtain the feature map Y_c of the channel attention weight, which is shown in (7):

$$Y_c = Y \otimes Z_c \quad (7)$$

The next step is to get the spatial attention weight Z_s after inputting Y_c to SAM. Z_s combines the channel and the spatial attention. In CBAM, Z_s is multiplied directly by the input feature Y. But in the construction environment, surveillance cameras are generally placed at the high places. The distance between the cameras and the construction personnel is far. It means that the detection objects are small, and the model needs to focus more on the target area.

Therefore, in this paper, the original is selected as the input feature map Y to focus on the target position, which means Y is multiplied by Z_s to obtain the feature map Y_s. It is shown as follows:

$$Y_s = Y \otimes Z_s \quad (8)$$

Y_s is obtained after inputting Y_c into SAM, and it contains the prior knowledge of the channel attention. Compared with Y_c, the weight Y_s focuses on more effective features of Y. In addition, the sum of Y_c and Y_s further fuses the features in the two dimensions of space and channel. Finally, sigmoid function is adopted to get the matrix Z_{cs} based on the channel spatial attention weights, which is shown in (9). The size of the weight reflects the attention position of the feature map, so that the model can obtain effective features more accurately.

$$Z_{cs} = \sigma(Y_c \oplus Y_s) \quad (9)$$

The parameter comparison after adding the improved BiCAM module to YOLOv5s is shown in Table 1. CBAM means that the CBAM attention module is embedded in the neck of the model, and BiCAM means that the BiCAM attention module proposed in this paper is embedded. It can be seen that both attention modules can improve the accuracy of model detection, and the effect of BiCAM is better. According to the analysis, BiCAM can extract broader semantic information and has better feature expression ability.

TABLE 1. Performance comparision after improving the attention module.

Module	Parameter	Map/%	Inference time/(ms.fps^{-1})
YOLOv5s	7.06	92.14	18
YOLOv5s+CBAM	7.09	93.02	18
YOLOv5s+BiCAM	7.11	93.49	18

IV. EXPERIMENT AND ANALYSIS

A. EXPERIMENTAL ENVIRONMENT AND EVALUATION INDEX

In the experiments, the model building, training and verification are all completed under the Pytorch framework, and CUDA is used to accelerate the calculation. The experimental configuration is shown in Table 2.

In this paper the open-source dataset safety helmet wearing dataset (SHWD) is selected to validate the helmet detection algorithm. SHWD includes various images, which cover different scenes, visual ranges, lighting, personal poses, and

TABLE 2. Experimental environment configuration.

Category	Entry	Version
Hardware Configuration	GPU	GeForce RTX 2080Ti
	CPU	AMD Ryzen 7 5800X
	Memory	32GB
Software Configuration	Syetem	Windows 10
	Python version	3.7
	Deep Learning Framework	Pytorch
	CUDA	10.0

occlusions. There are 7581 images in the dataset, including 9044 positive samples with helmets and 111514 negative samples without helmets. All the images are divided into training and validation sets at the ratio of 9:1, which means 6064 images are used for the training and 1517 images are used for the validation. Fig. 8 shows some training samples.



(a)Big object (b) Dense object (c) Contains negative sample objects

FIGURE 8. Partial training pictures.

The training of the model includes two stages, freeze training and unfreeze training. The first 50 epochs are freeze training, and the parameters of the backbone network are invariant. The last 100 epochs are for unfreeze training, and the parameters of the backbone network are variant. The image resolution is 640 × 640, the training batch is 16, the initial learning rate is 0.001, the IoU threshold is 0.5, and the model parameters are optimized by the adam optimizer.

In the experiment, the parameter quantity, recall rate, model size, average precision (mAP@.5) and inference time are used as the evaluation indicators of each model to validate the effectiveness of the YOLO-M model proposed in this paper.

Equation (10), (11) and (13) are the formulas for calculating recall rate, precision and mAP. *TP* is the true positive, which means how many positive samples are correctly identified. *FN* is the false negative, which means how many positive samples are not correctly identified. *FP* is the false positive, which means how many negative samples are incorrectly identified. Recall indicates how many positive examples in the sample are identified correctly, and precision indicates how many samples are identified correctly. In (12), $P(r)$ is the smoothed precision and recall curve, and AP is the area of the integral operation of $P(r)$. The mAP index is the average value of the AP values of all samples, as shown in (13). C is the number of classification, and $\mathrm{AP_i}$ is the accuracy of the i-th classification.

$$\mathrm{R} = \frac{TP}{TP + FN} \tag{10}$$

$$\mathrm{P} = \frac{TP}{TP + FP} \tag{11}$$

$$\mathrm{AP} = \int_0^1 P(r)dr \tag{12}$$

$$\mathrm{mAP} = \frac{\sum_{i=1}^{C} AP_i}{C} \tag{13}$$

The number of model parameters determines the size of the model, and it is also related to the memory usage during the model inference process. Too many parameters mean that the model is difficult to deploy, which is not conducive to the monitoring deployment of the construction site. It can be calculated as (14):

$$\text{Parameters} = c_o(k_w \times k_h \times c_i + 1) \tag{14}$$

In (14), c_o is the number of output channels of the current convolution layer, c_i is the number of input channels of the convolution layer, k_w is the width of the convolution kernel, k_h is the height of the convolution kernel, and the constant 1 means the bias.

B. TRAINING RESULTS

After training the YOLO-M model and the original YOLOv5s model for 150 epoches under the same configuration, the average precision (score_threhold=0.5) comparison curves are shown in Fig. 9, the average recall rate comparison curves are shown in Fig. 10, and the training Loss are shown in Fig. 11.

It can be concluded from the figures that the detection accuracy of the YOLO-M model proposed in this paper is better than that of YOLOv5s, which is improved by 2.22% under the premise of ensuring the basic recall.

C. EXPERIMENT COMPARISON

With the same training parameters, the experimental results of the average of precision, recall, number of parameters, mAP and inference time of YOLO-M, SSD, RetinaNet, Faster R-CNN, YOLOv3, YOLOv4, MobileNetv3-YOLOv4, YOLOv5s, YOLO-M (without BiCAM), YOLO-M (without ResFPN) and the excellent helmet wearing detection algorithms, YOLOX(ECA-Net) and MD-YOLO v2 are compared. The comparison of the effects of these models on helmet wearing detection is shown in Table 3.

From Table 3, compared with Faster RCNN, the model size of YOLO-M is reduced greatly, and the average accuracy is improved by 24.95%. Compared with the mainstream detection network models SSD and RetinaNet, the parameters of YOLO-M are greatly reduced, and the average accuracy

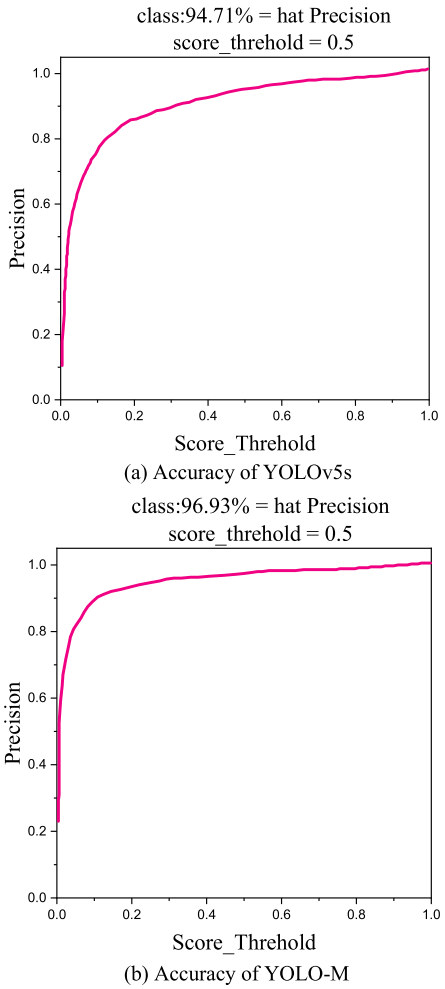


FIGURE 9. Model accuracy curves before and after improvement.

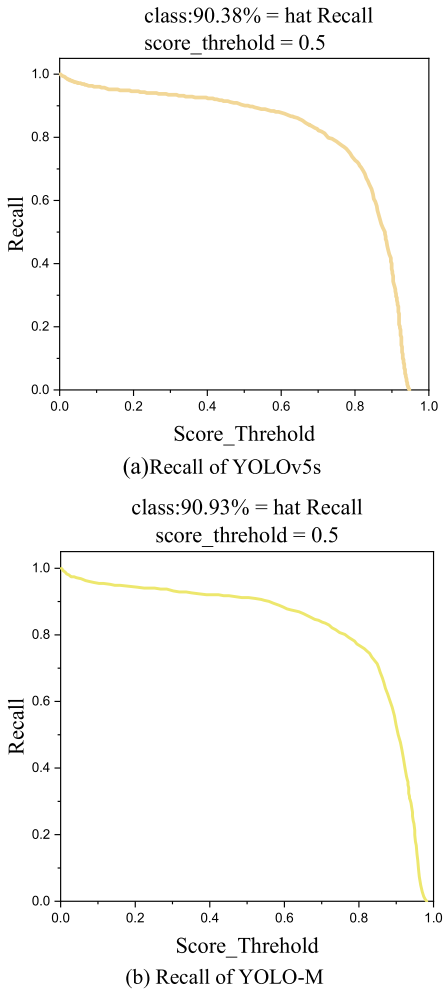


FIGURE 10. Model recall curves before and after improvement.

is improved. When compared with the YOLO series of networks, YOLO-M has the best performance, and the average accuracy is about 2% higher than that of YOLOv5s.

The average inference time of each image is 16ms, and it is 53ms, 31ms, 9ms, and 3ms faster than YOLOv3, YOLOv3-tiny, YOLOv4, and Mobilv3-YOLOv4 respectively. For the model parameters, it is 2M lower than YOLOv5s, but the accuracy is improved. When YOLO-M does not load the ResFPN feature fusion module, the model detection accuracy is 0.97% higher than that of YOLOv5s. When YOLO-M does not load the BiCAM attention module, the detection accuracy is 0.62% higher than that of YOLOv5s. The number of YOLO-M parameters is the least after modifying the backbone network, which reduces the computational cost. Considering the requirements of safety helmet detection accuracy, real-time performance in construction sites and the deployment cost of the model, YOLO-M performs the best.

In addition, it can be seen that the YOLO-M proposed in this paper is superior to the latest helmet detection researches YOLOX (ECA-Net) and MD-YOLO v2 on both mAP and Parameter. The mAP is 3.37% higher than YOLOX (ECA-Net), and 6.45% higher than MD-YOLO v2. The inference

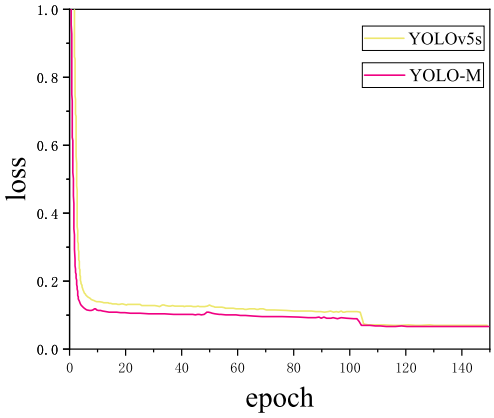


FIGURE 11. Loss curves.

time of the three methods is only different with a few milliseconds, and the model YOLO-M is sufficient to meet the requirements of helmet detection in construction sites.

To show the differences more effectively between the improved YOLO-M and YOLOv5s detection algorithms, small objects and objects occluded by steel bars are selected



(a) YOLOv5s for small objects (b) YOLO-M for small objects (c) YOLOv5s for occluded objects (d) YOLO-M for occluded objects

FIGURE 12. Comparison of detection effects.

TABLE 3. Comparison of the detection results of each algorithm.

Model	Precision/ %	Recall/%	mAP/%	Parameter/ M	Inference time/ (ms.fps^{-1})
SSD	83.55	46.71	65.68	26	10
RetinaNet	86.53	89.25	88.36	34.7	39
Faster R-CNN	71.98	76.79	73.48	137	36
YOLOv3	80.3	81.7	80.3	61.9	69
YOLOv4	90.4	88.7	89.7	64	25
MobileNetv3-YOLOv4	88.6	87.3	88.2	11	19
YOLOv5s	94.71	90.38	92.14	7.06	18
YOLO-M(without BiCAM)	95.33	89.76	92.27	5.3	17
YOLO-M(without ResFPN)	95.68	90.51	92.79	5.2	17
YOLOX(ECA-Net) [7]	-	-	90.5		14
MD-YOLO v2 [5]	-	-	87.42	17.5	8
YOLO-M	96.93	90.93	93.87	5.5	17

TABLE 4. Comparison of the detection results of filtered data sets.

Model	Average precision/(mAP) %	Recall/ %	Parameter/ M	Inference time/(ms.fps)$^{-1}$)
YOLO-M	96.10%	79.16%	5.5	16
YOLOv5s	93.87%	77.72%	7.06	18

for the comparison experiment. In Fig. 12 (a) and (b), YOLO-M can achieve better detection results for the small objects. Only 5 objects are identified in YOLOv5s, while 7 objects are identified in YOLO-M, and it detects the objects that YOLOv5s missed.

For the occluded objects in Fig. 11(c) and (d), the probabilities of two detected objects are 74% and 72% in YOLOv5s, while 76% and 83% in YOLO-M. It can be seen that YOLO-M maintains good detection accuracy and reduces the number of parameters.

Considering the complex environment and multi-target detection requirements in practical applications, 188 images of densely detected targets and with shadow occlusion in the dataset are selected. The results of the experiments are shown in Table 4. For the complex images, the model cannot mark the detection objects well. The recall of YOLOv5s is 77.72%, which decreases by 12.66%. The recall of YOLO-M is 79.16%, which decreases by 11.77%, but it is still better than the YOLOv5s model. At the same time, the accuracy of the two models decreases slightly. The accuracy of the YOLO-M model is 96.10%, which is 2.23% higher than that of YOLOv5s.

D. STABILITY ANALYSIS

In the above, the CBAM attention module and the Res-FPN feature fusion structure are used to improve the detection effect of YOLO-M. This section will analyze the stability of the YOLO-M model from regression box and training loss separately.

In the training process of the YOLO-M model, the accurate regression of the detection box is very important to obtain a stable model detection result. The early method of guiding the regression of the detection box is IOU loss. IOU is the overlap between the predicted box and the ground truth box, as shown in Fig. 13.

The calculation of IOU is shown as (15):

$$\mathrm{IOU} = \frac{A \cap B}{A \cup B} \tag{15}$$

However, IOU has two obvious disadvantages. When the two boxes do not intersect, the value of IOU will be 0, which will make it impossible to calculate the transfer gradient. When the two boxes are contained, the relative position of the two boxes cannot be determined. In addition, the slow

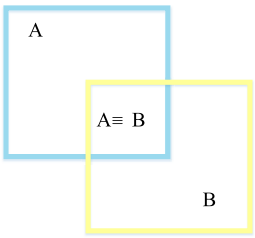


FIGURE 13. IOU diagram.

FIGURE 14. IOU diagram.

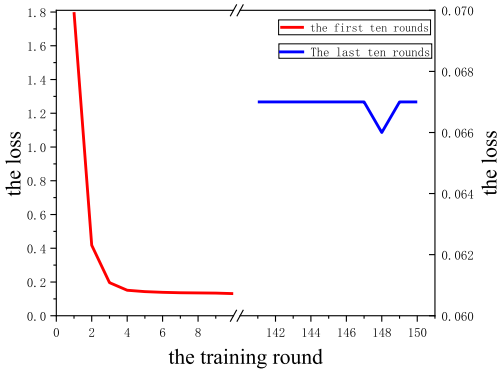


FIGURE 15. The training loss records.

convergence speed of IOU loss gradually fails to meet the performance requirements.

To filter out more accurate prediction boxes to improve the stability of the model, in this paper CIOU loss is used to calculate the loss value of the detection box and the ground truth box. The calculation is shown as (16), (17) and (18):

$$L_{CIOU} = 1 - IOU + \frac{\rho^2\left(b, b^{gt}\right)}{c^2} + \alpha v \tag{16}$$

$$v = \frac{4}{\pi^2}(arctan\frac{w^{gt}}{h^{gt}} - arctan\frac{w}{h})^2 \tag{17}$$

$$a = \frac{v}{(1 - IOU) + v} \tag{18}$$

In the above formulas, ρ^2 represents the square of the Euclidean distance between the center points of the two boxes, c^2 is the square of the diagonal distance of the outer box when the other box is included. The values of w^{gt} and h^{gt} represent the width and the height of the ground truth box separately. The value of α is the weight coefficient, and v is used to measure the similarity of the length-width ratio between the prediction box and the ground truth box.

By introducing the distance between the center points and the distance between the diagonals, the inclusion of the two boxes can be effectively solved, as shown in Fig. 14.

When inclusion occurs, the value of IOU is 0, but the distance between the center points of the two boxes is not 0, so the gradient can be calculated for backpropagation.

Then this paper analyzes the stability of YOLO-M from the training process. At the beginning of training, the model loss value drops sharply, as shown in the red curve in Figure 15. After multiple rounds of training, the loss value stabilizes at 0.067, as shown by the blue curve. It shows that the model has good stability after 150 rounds of training on the SHWD dataset.

V. CONCLUSION

In this paper, for the problems of the existing helmet wearing detection model on too many parameters, high hardware cost, and poor recognition of occluded targets and distant dense targets, a helmet wearing detection model YOLO-M is proposed. The experimental results show that the improved YOLO-M model maintains high detection accuracy and has a faster detection speed. However, the construction scenes are changeable, and there are still cases of missed detection in complex scenes. In the future, we will focus on the robustness of the model on the various scenarios.

REFERENCES

[1] Q. Y. Li, J. B. Wang, and H. W. Wang, "Study on impact resistance of industrial safety helmet," *J. Saf. Sci. Technol.*, vol. 17, no. 3, pp. 182–186, Mar. 2021, doi: 10.11731/j.issn.1673-193x.2021.03.028.

[2] Y. X. Wang, Z. Wang, and B. Wu, "Research review of safety helmet wearing detection algorithm in intelligent construction site," *J. Wuhan Univ. Technol.*, vol. 43, no. 10, pp. 56–62, Oct. 2021, doi: 10.3963/j.issn.1671-4431.2021.10.00.

[3] L. Jun, W. C. Dang, and P. Lihu, "Safety helmet detection based on YOLO," *Comput. Syst. Appl.*, vol. 28, no. 9, pp. 174–179, Sep. 2019, doi: 10.15888/j.cnki.csa.007065.

[4] W. Bing, L. Wenjing, and T. Huan, "Improved YOLOv3 algorithm and its application in helmet detection," *Comput. Eng. Appl.*, vol. 26, no. 9, pp. 33–40, Feb. 2020, doi: 10.3778/j.issn.1002-8331.1912-0267.

[5] F. Ming, S. Tengteng, and S. Zhen, "Fast helmet-wearing-condition detection based on improved YOLOv2," *Opt. Precis. Eng.*, vol. 27, no. 5, pp. 1196–1205, Mar. 2019, doi: 10.3788/OPE.20192705.1196.

[6] H. C. Zhao, X. X. Tian, and Z. S. Yang, "YOLO-S: A new lightweight helmet wearing detection model," *J. East China Normal Univ. Natural Sci.*, vol. 47, no. 5, pp. 134–145, Sep. 2021, doi: 10.3969/j.issn.1000-5641.2021.05.012.

[7] T. Ding, X. Y. Chen, Q. Zhou, and H. L. Xiao, "Real-time detection of helmet wearing based on improved YOLOX," *Electron. Meas. Technol.*, vol. 45, no. 17, pp. 72–78, Sep. 2022, doi: 10.19651/j.cnki.emt.2209425.

[8] X. Ma, K. Ji, B. Xiong, L. Zhang, S. Feng, and G. Kuang, "Light-YOLOv4: An edge-device oriented target detection method for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10808–10820, 2021, doi: 10.1109/JSTARS.2021.3120009.

[9] Z. Z. Sun, X. G. Len, and L. Yu, "BiFA-YOLO: A novel YOLO-based method for arbitrary-oriented ship detection in high-resolution SAR images," *Remote Sens.*, vol. 13, no. 21, pp. 4209–4237, Oct. 2021, doi: 10.3390/rs13214209.

[10] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

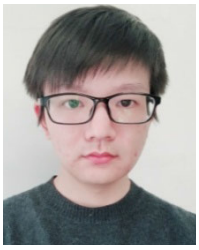
[11] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[12] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[13] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The PASCAL visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Jun. 2010.
[14] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.
[15] W. Liu, D. Anguelov, and D. Erhan, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Oct. 2016, pp. 21–37.
[16] A. Howard, M. Sandler, B. Chen, W. Wang, L.-C. Chen, M. Tan, G. Chu, V. Vasudevan, Y. Zhu, R. Pang, H. Adam, and Q. Le, "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1314–1324.
[17] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[18] Y. H. Shao, D. Zhang, and H. Y. Chu, "A review of Yolo object detection based on deep learning," *J. Electron. Inf. Technol.*, vol. 44, no. 10, pp. 3697–3708, Oct. 2022, doi: 10.11999/JEIT210790.
[19] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 658–666.
[20] A. Neubeck and L. Van Gool, "Efficient non-maximum suppression," in *Proc. 18th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2006, pp. 850–855.
[21] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.
[22] S. Woo, J. Park, and J. Y. Lee, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.



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